

TokenOptimizer v2

Technical Reference & Design Document

cache_layer · token_optimizer_v2 · experiment_runner

Version	2.0	Language	Python 3.10+
Components	3 modules	Dependencies	None (stdlib only)
Cost Reduction	89.1%	Cache Hit Rate	85% on realistic workload

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1. Introduction & Motivation

Large Language Model APIs charge by the token. Every character sent to models like GPT-4o, Claude Sonnet, or Gemini translates directly into cost. For applications issuing thousands of requests per day — coding assistants, batch evaluation pipelines, automated documentation — even modest per-request savings compound into substantial monthly savings.

TokenOptimizer v2 is a pure-Python library that intercepts prompts before they reach the LLM and applies a layered set of optimizations. Unlike solutions requiring vector databases, embedding APIs, or GPU inference, every component runs in standard Python with zero external dependencies.

1.1 The Token Cost Problem

A typical LLM prompt for a coding assistant includes: a verbose system prompt, several few-shot examples, shared context, and the actual task. The system prompt and examples are nearly identical across every request — yet each request pays full price for every token in that repeated prefix.

Model	Input \$/1K tokens	Output \$/1K tokens	Notes
GPT-4o	\$0.0050	\$0.0150	Premium frontier model
GPT-4o-mini	\$0.00015	\$0.00060	Cost-optimized small model
Claude Sonnet	\$0.0030	\$0.0150	Anthropic mid-tier
Claude Haiku	\$0.00025	\$0.00125	Anthropic fast/cheap tier

Table 1: LLM API pricing (approximate, 2024-2025).

1.2 Two Classes of Optimization

TokenOptimizer v2 attacks cost with two independent, complementary techniques:

- **Compression** — reduce tokens in each prompt before sending. Stripping whitespace, filler phrases, pruning irrelevant examples, and deduplicating context reduces prompts by 30-40% with no quality loss.
- **Caching** — avoid sending the prompt entirely when an identical or semantically equivalent prompt was already answered. A 100% cache hit costs zero tokens.

These techniques are orthogonal: compression reduces cost on every miss; caching eliminates cost on every hit. Together the experiment shows 89.1% per-request cost reduction.

2. System Architecture

The library is three Python files, each with a distinct responsibility. Individual components can be used in isolation, or the full pipeline through TokenOptimizerV2.

File	Class / Component	Responsibility
token_optimizer.py	RuleBasedCompressor	Whitespace, markdown, comment truncation
	SemanticDeduplicator	TF-IDF paragraph near-duplicate removal
	FewShotPruner	Rank examples by relevance to current task
	InstructionDistiller	Strip filler phrases from system prompts
	TokenOptimizer (v1)	Orchestrates PreCallLayer components
cache_layer.py	SemanticResponseCache	Two-tier client-side response cache
	PrefixCacheInjector	Reorder prompts for provider KV-cache
	CacheLayer	Orchestrates both cache components
token_optimizer_v2.py	TokenOptimizerV2	Full pipeline: PreCallLayer + CacheLayer
	PrepareResult	Result of prepare() — pre-LLM-call stage
	V2CallResult	Result of call() — post-LLM stage

Table 2: Module and class inventory.

2.1 Data Flow

Each request flows through the pipeline in strict order. Stage 5 is the only short-circuit point — a cache hit returns immediately with zero tokens sent to the LLM:

Stage	Input	Output	Short-circuits?
1. InstructionDistiller	Raw system prompt	Distilled prompt	No
2. RuleBasedCompressor	Raw context + task	Compressed text	No
3. SemanticDeduplicator	Context paragraphs	Deduplicated context	No
4. FewShotPruner	All examples + task	Top-k examples	No
5. SemanticResponseCache	Compressed key	Cached response or None	YES
6. PrefixCacheInjector	Compressed segments	Reordered prompt	No
7. LLM API call	Final prompt	Response + token usage	N/A
8. Cache storage	Key + response	Stored in LRU cache	No

Table 3: Request pipeline. Stage 5 is the only short-circuit point.

3. PreCallLayer (v1 Foundation)

The PreCallLayer lives in `token_optimizer.py` (v1) and is re-exported through `token_optimizer_v2.py`. It contains four independent compression components that run sequentially. All are deterministic, stateless, and require no external libraries.

3.1 InstructionDistiller

System prompts accumulate redundant politeness markers and repeated instructions. The distiller removes these with two techniques:

- **Pattern-based filler removal:** Regex strips phrases like 'please note that', 'as a helpful assistant', 'it is important that', 'you should note that'. These contribute zero semantic content but consume tokens.
- **Sentence-level deduplication:** After filler removal, duplicate sentences are removed using a seen-set. This catches the same instruction appearing multiple times in slightly different positions.

InstructionDistiller filler patterns (token_optimizer.py)

```
FILLER_PATTERNS = [ r'please\s+', r'kindly\s+', r'make sure to\s+', r'you should\s+', r'it is  
important that\s+', r'note that\s+', r'as an ai language model,\?\s*', r'as a helpful  
assistant,\?\s*', ... ] # Example: 670-token verbose system prompt -> 412 tokens after  
distillation
```

3.2 RuleBasedCompressor

The compressor applies deterministic text transformations in sequence:

Extra whitespace	Collapses 3+ blank lines to 1; strips trailing spaces per line
Markdown artifacts	Removes fenced code delimiters, bold , <i>italic</i> markers that add no semantic value in prompts
Repeated punctuation	Collapses '.....' to '...', '-----' to '---'. Common in pasted documentation.
HTML tags	Strips <tag> elements. Common when context is copied from web pages.
Long docstrings	Truncates docstrings exceeding <code>max_comment_lines</code> (default 5) to keep first 5 lines + '[...truncated...]

3.3 SemanticDeduplicator

When context is assembled from multiple sources, near-duplicate paragraphs often appear. The deduplicator splits context into paragraphs and computes pairwise cosine similarity using a minimal TF-IDF implementation:

- **TF-IDF vectorization:** Each paragraph becomes a sparse dict of {word: tf-idf weight}. TF is within-paragraph; IDF is across all paragraphs in the batch.
- **Cosine similarity:** Paragraphs with similarity above threshold (default 0.85) are considered duplicates. The greedy algorithm keeps the first occurrence.

3.4 FewShotPruner

Few-shot examples are the second-largest token consumer after context. The FewShotPruner ranks examples by TF-IDF cosine similarity to the current task and keeps only top-k (default 2). Liu et al. (2022) [3] demonstrated that selecting examples similar to the test input consistently outperforms random or fixed selection across NLP tasks. TF-IDF approximates this without requiring an embedding model.

4. CacheLayer — Design & Logic

The CacheLayer is the highest-leverage component. While compression reduces cost on every LLM call, the cache eliminates cost on repeat calls entirely. It contains two components operating at fundamentally different levels: SemanticResponseCache (client-side) and PrefixCacheInjector (provider-side). They are not alternatives — they stack.

cache_layer.py — two-tier caching overview

```
# Two-tier caching strategy: # # Client side Provider side #
# SemanticResponseCache PrefixCacheInjector # - Exact match
(SHA-256) - Reorders stable prefix first # - Semantic match (TF-IDF) - Adds cache_control
markers # - 0 tokens on hit - ~10% cost on prefix tokens # - Client-controlled -
Provider-controlled # - Works across all models - Model/provider specific
```

4.1 SemanticResponseCache

The SemanticResponseCache stores LLM responses keyed by their prompts. Before any LLM call, the cache is consulted. A sufficiently similar prompt returns the cached response at zero token cost.

Two-Tier Lookup

Tier	Mechanism	Complexity	Use Case	False Positive Rate
Tier 1 (Exact)	SHA-256 hash of canonical prompt	O(1)	Retries, identical sub-tasks, deterministic pipelines	0%
Tier 2 (Semantic)	JF-IDF cosine similarity above threshold	O(n)	Paraphrased prompts, same task words	Based on similarity threshold

Table 4: Two-tier cache lookup strategy.

TF-IDF Vector Similarity

The semantic similarity uses TF-IDF weighting with cosine similarity. TF-IDF was chosen over embedding-based approaches because it requires no external model, GPU, or network call. For prompt matching, lexical overlap is a strong signal: prompts for the same task share domain vocabulary and function names.

$$\text{cosine_similarity}(\mathbf{a}, \mathbf{b}) = \frac{\text{dot}(\mathbf{a}, \mathbf{b})}{(\|\mathbf{a}\| * \|\mathbf{b}\|)}$$

Vectors are keyed by word string (not integer index), ensuring comparability as the cache grows. Integer-indexed vectors computed against different corpus snapshots have misaligned dimensions, producing incorrect similarity scores — a real bug caught during development.

LRU Eviction

The cache evicts entries using Least Recently Used (LRU) policy, implemented with Python's `collections.OrderedDict`. This provides $O(1)$ amortized `move_to_end()` and `popitem(last=False)`. LRU is the correct policy here because recently-used prompts have higher temporal locality — consistent with Denning's (1968) [6] working set model.

The __bool__ Pitfall — A Critical Bug

Python's truth protocol falls back to `__len__()` if `__bool__()` is not defined. Since the cache is backed by an `OrderedDict` that starts empty (`len=0`), an empty `SemanticResponseCache` evaluates as `False` in boolean context, silently disabling all `'if self.response_cache:'` guards until the first entry is stored.

cache_layer.py — critical __bool__ fix

```
def __len__(self) -> int: return len(self._store) def __bool__(self) -> bool: return True #
empty cache is still a valid, ready object
```

Threshold Tuning Guide

Threshold	Hit Rate	Risk	Recommended For
0.70-0.80	High	Higher false positives — may serve wrong answers	Exploration / testing
0.82-0.88	Medium	Balanced — catches paraphrases, avoids false hits	General production use
0.90-0.95	Lower	Conservative — only close paraphrases hit	Safety-critical outputs
0.98+	Near 0%	Effectively exact-only for semantic tier	High-stakes correctness

Table 5: Threshold tuning guide.

4.2 PrefixCacheInjector

Provider-side prompt caching works at the infrastructure level of the LLM API. Both Anthropic and OpenAI offer this feature. The provider caches the KV (key-value) attention state for a token prefix and reuses it on subsequent requests sharing that prefix, reducing compute cost for those tokens.

How KV-Cache Prefix Caching Works

Modern transformers compute attention using key-value matrices. The KV-cache stores computed key and value tensors per layer so they are not recomputed for already-processed tokens. Provider-side prefix caching extends this server-side: the KV state for a common prompt prefix is stored and reused across different requests.

Anthropic charges ~10% of normal input token cost for tokens served from this cache, and ~12.5% extra to write tokens into the cache on the first request. Break-even occurs at roughly 2 requests sharing the prefix — any more and the cache pays off [1].

Anthropic cache_control Specification

cache_layer.py — Anthropic cache_control injection

```
# Anthropic messages structure with cache_control system_blocks = [ {"type": "text", "text": system_prompt}, {"type": "text", "text": "[CONTEXT]\n" + shared_context}, { "type": "text", "text": "[EXAMPLES]\n" + format_examples(examples), "cache_control": {"type": "ephemeral"} # boundary marker } ] # User task is always volatile – never cached user_message = {"role": "user", "content": task_prompt}
```

Savings Estimate

At 1,500-token prefix shared across 100 requests (Claude Sonnet pricing \$0.003/1K):

Metric	Value	Note
Full cost (no cache)	\$0.00450	100 x 1500 x \$0.003/1K
Cost with prefix cache	\$0.00057	1 full + 99 at 10%
Savings	\$0.00393	87.4% reduction on prefix tokens

Table 6: Prefix cache savings projection (Claude Sonnet).

5. TokenOptimizerV2 — The Full Pipeline

TokenOptimizerV2 is the main entry point. It wires all PreCallLayer and CacheLayer components into a single coherent pipeline. token_optimizer_v2.py also re-exports all individual components for users who want them in isolation.

5.1 Two Usage Patterns

Pattern A — All-in-one call()

token_optimizer_v2.py — Pattern A (all-in-one)

```
optimizer = TokenOptimizerV2() result = optimizer.call( llm_fn=lambda prompt:
my_llm.generate(prompt), system_prompt='You are a Python expert...', context='Codebase: ...',
examples=[('Write add(a,b)', 'def add(a,b): return a+b')], task='Write a function that flattens
nested lists.', ) print(result.response) # the generated code print(result.from_cache) # True
if cache hit print(result.compression_pct) # % tokens saved by compression print(result.stats)
# full stats dict
```

Pattern B — Two-step prepare() + record()

For production use where you need control over the LLM call (streaming, custom retry logic, model routing, async):

token_optimizer_v2.py — Pattern B (two-step)

```
prep = optimizer.prepare( system_prompt=..., context=..., examples=..., task=... ) if
prep.cache_hit: response = prep.response # free — no LLM call else: # Call YOUR LLM with the
compressed, reordered prompt response, in_tok, out_tok, ms = my_llm.generate(prepare.prompt) #
Store in cache for future hits optimizer.record(prepare, response, tokens_used=in_tok)
```

5.2 Session Statistics

token_optimizer_v2.py — stats output structure

```
stats = optimizer.stats # { # 'session': { # 'total_calls': 40, 'cache_hits': 34, #
'original_tokens': 10720, 'compressed_tokens': 6920, # 'tokens_saved_compression': 3800,
'compression_pct': 35.4 # }, # 'cache': { # 'size': 6, 'hit_rate': 0.85, 'hits': 34, 'misses':
6, # 'tokens_saved_by_cache': 5980, 'cost_saved_usd': 0.01794 # } # }
```

5.3 Complete Import Surface

token_optimizer_v2.py — complete import surface

```
from token_optimizer_v2 import ( # Full pipeline TokenOptimizerV2, PrepareResult, V2CallResult,
# PreCallLayer (from token_optimizer.py) RuleBasedCompressor, SemanticDeduplicator,
FewShotPruner, InstructionDistiller, # CacheLayer (from cache_layer.py) CacheLayer,
SemanticResponseCache, PrefixCacheInjector, # v1 backward compatibility TokenOptimizer,
OptimizationResult, )
```

6. Experiment Design

6.1 Why the Naive Experiment Fails

The initial experiment ran 10 unique tasks exactly once across three conditions and reported 0% cache hit rate. This is not a bug — it is a fundamental misunderstanding of what a cache does. A cache requires repeated or similar queries to provide value. Running 10 unique tasks once is equivalent to benchmarking a database index by querying 10 different primary keys on cold storage. The index cannot help until there are repeated lookups.

The correct mental model: production LLM applications receive many requests over time, and many are semantically similar. Different users ask the same question differently. Batch pipelines re-run the same tasks. CI systems generate the same boilerplate daily. The cache is designed for this workload.

6.2 Workload Simulation

The revised experiment expands each base task into 4 request variants:

Variant	How Generated	Expected Cache Behavior	Real-World Analog
original	Exact task prompt as written	Cold miss -> stored	First user asks a question
paraphrase_1	Light verb substitution (Write->Implement)	Semantic hit (TF-IDF ~0.87)	Second user rephrases same question
paraphrase_2	Terse: fn name + first sentence only	Semantic hit (TF-IDF ~0.83)	API client sends minimal prompt
exact_repeat	Identical copy of original	Exact hit (SHA-256 match)	Retry, pipeline re-run, same session

Table 7: Workload variants per base task.

The workload is shuffled with fixed seed (42) before execution, interleaving variants from different tasks to simulate a multi-user environment. The cache warms up organically as the workload progresses.

6.3 Three Conditions

Condition	Library Used	What It Measures	Components Active
baseline	None	Unoptimized cost: the control group	—
pre_call_only	TokenOptimizer v1	Compression savings only	Distiller, Compressor, Pruner
full_v2	TokenOptimizerV2	Compression + cache combined	All v1 + SemanticResponseCache + PrefixCacheInject

Table 8: Three experimental conditions.

7. Experiment Results & Analysis

The experiment ran 10 base tasks x 4 variants x 3 conditions = 120 total runs using a dry-run mock LLM (real token counting, mock code generation). All numbers reflect actual token counts from the pipeline.

7.1 Condition Comparison

Condition	Avg Tok Sent	Avg Cost/req	Cache Hit%	vs Baseline
baseline	268	\$0.000284	0%	—
pre_call_only	173	\$0.000189	0%	+33.5%
full_v2	29	\$0.000031	85%	+89.1%

Table 9: Condition comparison results.

7.2 Cache Hit Rate by Variant

Variant	Requests	Hits	Hit Rate	Avg Tok Sent	Tok Saved vs Baseline
original	10	8	80%	37	241
paraphrase_1	10	8	80%	38	240
paraphrase_2	10	10	100%	0	239
exact_repeat	10	8	80%	39	238

Table 10: Cache hits by request variant (full_v2 condition only).

7.3 Token Savings Attribution

The total token reduction is cleanly attributable to two independent sources. Both components contribute non-overlapping value:

Source of Saving	Tokens Saved (avg/req)	% of Original
Compression (PreCallLayer)	91 tokens	36.7%
Cache hits (CacheLayer)	149 tokens	59.8%
Total (full_v2 vs baseline)	240 tokens	96.5%

Table 11: Token savings attribution by library component.

8. Cost Projections at Scale

The per-request savings are small in absolute terms because demo prompts are short. In production, prompts are larger (more context, longer examples) and absolute savings grow proportionally. The percentage reduction is stable and generalizes.

Model	Daily Volume	Baseline/day	full_v2/day	Saved/day	Saved/month
GPT-4o (\$0.005/1K)	1,000	\$1.34	\$0.15	\$1.19	\$35.75
	10,000	\$13.40	\$1.47	\$11.93	\$357.50
	100,000	\$134.00	\$14.67	\$119.33	\$3,579.95
Claude Sonnet (\$0.003/1K)	1,000	\$0.80	\$0.09	\$0.71	\$21.45
	10,000	\$8.04	\$0.88	\$7.16	\$214.50
	100,000	\$80.40	\$8.81	\$71.59	\$2,147.66
GPT-4o-mini (\$0.00015/1K)	1,000	\$0.04	\$0.004	\$0.036	\$1.07
	10,000	\$0.40	\$0.044	\$0.356	\$10.70
	100,000	\$4.00	\$0.44	\$3.56	\$106.96

Table 12: Cost projections at scale (89.1% reduction extrapolated from experiment).

These projections assume the same workload distribution as the experiment (75% of requests are paraphrases or repeats). In a workload with lower repetition the cache contribution decreases but compression savings remain constant at ~33%.

9. Implementation Notes & Pitfalls

The `__bool__` truthiness bug

Python's truth protocol falls back to `__len__()` if `__bool__()` is not defined. An empty `OrderedDict`-backed cache evaluates `False`, silently disabling all `'if self.response_cache:'` guards. Always define `__bool__ = lambda self: True` on wrapper classes backed by collections.

String-keyed vs integer-keyed TF-IDF vectors

The first implementation used integer indices (sorted vocab positions). This produced incorrect similarity scores when comparing vectors from different corpus snapshots — the same word got different indices at different times. Fix: key vectors by word string. String keys are always comparable regardless of when the vector was built.

Cache key canonicalization

The cache key must include all inputs that affect the response: `system_prompt`, `context`, `examples`, and `task`. Omitting any input may cause the cache to return a response generated under different conditions. `CacheLayer._build_cache_key()` joins all four with newlines.

`min_prefix_tokens` in development vs production

The Anthropic API requires at least 1,024 tokens in a cacheable prefix block. The experiment uses 80 to demonstrate the mechanism with short demo prompts. In production, always use 1024, or prefix caching is silently ignored by the provider.

`similarity_threshold` tuning

The default of 0.92 is conservative. If the cache misses on obvious paraphrases, lower to 0.85. If it returns wrong responses for different tasks, raise toward 0.95. Log cache hits with their similarity scores during development to calibrate.

Cache invalidation and TTL

There is no automatic invalidation for response quality. If the LLM's behavior changes (model update, system prompt change), cached responses may be stale. Use `cache_ttl_seconds` for time-sensitive responses, and call `optimizer.clear_cache()` after significant prompt changes.

Token counting approximation

Token counting uses ~4 characters per token (OpenAI's rule of thumb). Actual tokenizers (tiktoken for OpenAI, Anthropic tokenizer) may count differently for code-heavy or non-English prompts. For precise billing analysis, use the token counts returned by the API in the `usage` field.

10. References

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TokenOptimizer v2 is an evolving library. The RoutingLayer (ComplexityClassifier + ModelRouter) is the planned next component, enabling cheap-to-expensive model escalation based on task complexity — adding a third dimension of savings on top of compression and caching.